

6(a)	similarity: lines are radial / greater separation of lines with increased distance from the sphere	B1
	difference: gravitational lines directed towards sphere <u>and</u> electric lines directed away from sphere	B1
6(b)(i)	$E = Q / 4\pi\epsilon_0 r^2$ or $E = kQ / r^2$ with k defined / substituted in	C1
	$4.1 \times 10^{-5} = [Q / (4\pi \times 8.85 \times 10^{-12} \times 0.025^2)] - [Q / (4\pi \times 8.85 \times 10^{-12} \times 0.075^2)]$	C1
	$Q = 3.2 \times 10^{-18} \text{ C}$	A1
6(b)(ii)	smooth curve with gradient decreasing starting at $(0, 4.1 \times 10^{-5})$ to d -axis at $(2.5, 0)$	B1
	smooth curve with gradient increasing from $(2.5, 0)$ ending at $(5, -4.1 \times 10^{-5})$	B1
6(b)(iii)	acceleration decreases (to zero at mid-point)	B1
	then acceleration increases in the opposite direction / increasing negative acceleration	B1

Question	Answer	Marks
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